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Case 14-2024: A 30-Year-Old Woman with Back Pain, Leg Stiffness, and Falls

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PRESENTATION OF CASE

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CME



Dr. Alba Coraini (Neurology): A 30-year-old woman was evaluated in the neurology clinic of this hospital because of back pain and leg stiffness.

The patient had been in her usual state of health until 3 years before the current presentation, when stiffness in the back and upper legs developed abruptly while she attempted to rise from a seated position. She also began to have associated low back pain that worsened with bending the knees or climbing stairs. During the next several months, back and leg stiffness waxed and waned; at times, the patient could walk normally and engage in running for exercise, and at other times, she was unable to walk because her knees felt as though they had “locked up.” Two months after the onset of symptoms, the patient fell and broke her right arm. She perceived the fall as being unusual in that after she tripped, she was unable to prevent herself from falling because of tension in her legs. After the fall, the patient underwent physical therapy for several months, but she did not resume running for exercise because of a fear of falling.

Two and a half years before the current presentation, the patient was evaluated in a rheumatology clinic of another hospital. Passive flexion of the left knee was guarded, but the knee bent when a distraction technique was used; the remainder of the examination was normal. Hip and knee radiographs were reportedly unremarkable. The knee stiffness was not thought to be due to an underlying rheumatologic disorder.

One and a half years before the current presentation, the patient was evaluated for knee pain in the sports medicine clinic affiliated with this hospital. The pain was localized to the anterior portions of the knees at the level of the patellae; it worsened with climbing stairs and resolved with rest. Examination and radiographs of the knees showed no abnormalities. Patellofemoral pain syndrome was diagnosed, and physical therapy was recommended. The knee pain resolved after physical therapy, but the patient continued to have leg stiffness and did not return to running owing to a fear of falling.

Three months before the current presentation, the patient noticed ongoing

lower back pain and increased stiffness in the legs that was worse in the right leg. She felt unsteady on her feet and worried about tripping while walking or falling while climbing stairs. While walking in her house, she hit her leg on a coffee table; both legs stiffened, and she fell, striking her face on the floor. There was no loss of consciousness. She was evaluated by her primary care physician. On examination, she had mild tenderness on palpation of the lumbar spine. Imaging studies were obtained.

Dr. Pamela W. Schaefer: Magnetic resonance imaging (MRI) of the lumbar spine, performed without the administration of intravenous contrast material, showed normal paraspinal soft tissues and exaggeration of the normal lumbar lordosis, with preserved vertebral height. A loss of normal signal intensity was noted in the L4–L5 intervertebral disk space on T2-weighted images, a finding that was consistent with mild degeneration. Evaluation for signal abnormality in the visible portion of the spinal cord and in the cauda equina nerve roots could not be assessed owing to severe motion artifact. There was no evidence of high-grade spinal canal or foraminal stenosis.

Dr. Coraini: Treatment with cyclobenzaprine, which was to be taken as needed at night for back pain, was initiated, and physical therapy was recommended. However, after 3 months of physical therapy, the back pain had not abated, and the leg stiffness had worsened. The patient sought evaluation in the neurology clinic of this hospital.

In the neurology clinic, the patient reported ongoing leg stiffness and rigidity that caused gait abnormalities. She noticed jerking movements at night as she was falling asleep. Three months before the current evaluation, she had taken lorazepam for claustrophobia before entering the MRI machine for spinal imaging, and her gait was reportedly normal for several hours after imaging.

The patient had a history of vitiligo, eczema, and alopecia areata. Graves' disease, for which she had been treated with methimazole, was in remission. Immune thrombocytopenia, which had been diagnosed 7 years before the current presentation, had been treated with a course of intravenous immune globulin. In addition to cyclobenzaprine, medications included cetirizine, desogestrel–ethinyl estradiol, and a multivitamin.

She had no known allergies. She lived in New England with her husband and had recently started a new job as an office manager. She was a lifelong nonsmoker and did not drink alcohol or use illicit drugs. One sister had autoimmune thyroiditis, and another had alopecia areata.

On examination, the temporal temperature was 36.6°C, the blood pressure 141/91 mm Hg, the pulse 100 beats per minute, the respiratory rate 16 breaths per minute, and the oxygen saturation 98% while the patient was breathing ambient air. The body-mass index (the weight in kilograms divided by the square of the height in meters) was 21.7. The patient appeared anxious but was in no acute distress. Strength was assessed as normal in the arms and legs. She had mildly increased tone in the legs; no fasciculations were present. Deep-tendon reflexes were 2+ in the arms and 3+ in the legs, with nonsustained clonus in the ankles. An exaggerated startle response was present. Sensation was normal. On ambulation, there was reduced flexion of the knees and a wide-based gait.

Blood levels of electrolytes were normal, as were the results of tests of kidney function. The blood level of creatine kinase was 33 U per liter (reference range, 26 to 192), the C-reactive protein level 1 mg per liter (reference range, 0 to 10), and the erythrocyte sedimentation rate 2 mm per hour (reference range, 0 to 20). Imaging studies were obtained.

Dr. Schaefer: MRI of the thoracic and lumbar spine, performed before and after the administration of intravenous contrast material (Fig. 1), was limited by motion artifact. No specific cord abnormality was identified. Mild degenerative changes were noted in the thoracolumbar spine, without moderate or severe narrowing of the spinal canal or neural foramina. No abnormal enhancement was seen. Possible atrophy of the posterior paraspinal musculature was detected in the lower lumbar spine. Exaggeration of the normal lumbar lordosis was noted.

Dr. Coraini: A diagnostic test was performed.

DIFFERENTIAL DIAGNOSIS

Dr. Christopher T. Doughty: This 30-year-old woman began to have low back pain with associated stiffness when she was 27 years of age. Low back pain is a common presenting symptom and is often self-limited. However, this patient also

described disabling stiffness that affected muscles in the legs and led to falls. Orthopedic and rheumatologic conditions can lead to stiffness of the affected joints; this patient was evaluated by both a physiatrist (a specialist in physical medicine and rehabilitation) and a rheumatologist during her illness, but no diagnosis was made. Although she initially described low back stiffness, ankylosing spondylitis is unlikely, given the lack of abatement with exercise, the normal C-reactive protein level and erythrocyte sedimentation rate, the unremarkable radiographs and MRIs of the lumbar spine, and the development of prominent leg stiffness.¹ In working toward the most likely diagnosis in this case, I will consider neurologic conditions associated with muscle stiffness (Table 1).

PERIPHERAL NERVOUS SYSTEM DISORDER

Most peripheral nervous system disorders associated with muscle symptoms cause muscle weakness rather than muscle stiffness. Myotonic disorders are notable exceptions. Myotonia refers to a failure of muscle relaxation after voluntary activation that patients perceive as stiffness.

On examination, myotonia can appear as delayed eye opening after forcible eye closure or as delayed hand opening after making a fist. Percussion myotonia can be elicited by striking a muscle with a reflex hammer and then observing sustained muscle contraction.² Myotonia congenita and paramyotonia congenita are the most common forms of nondystrophic myotonia. Both are genetic channelopathies that cause muscle stiffness with or without weakness.^{3,4} Myotonia congenita is associated with a “warm-up” phenomenon, in which muscle stiffness abates with ongoing activity. In persons with paramyotonia congenita, repeated exercise or cold temperature worsens stiffness. This patient’s symptoms did not follow the typical pattern of either myotonia congenita or paramyotonia congenita. In addition, no myotonia was identified on examination. Moreover, both conditions typically manifest in childhood.

Peripheral nerve hyperexcitability syndromes such as cramp-fasciculation syndrome and neuro-myotonia can be associated with muscle stiffness and are usually brought on by voluntary muscle activity.^{5,6} The hallmark of these disorders, however, is the presence of generalized

Table 1. Disorders Associated with Episodic Muscle Stiffness.

Disorder	Triggers for Stiffness	Alleviating Factors	Key Neurologic Examination Findings
Myotonia congenita	Prolonged rest before movement Cold temperature	Ongoing activity (“warm-up” phenomenon)	Generalized muscular hypertrophy Action myotonia Percussion myotonia
Paramyotonia congenita	Repeated exercise Cold temperature	Warm temperature	Generalized muscular hypertrophy Action myotonia
Peripheral nerve hyperexcitability	Activity	Rest	Fasciculations Myokymia
Dystonia	Varies with specific syndrome; some are task-specific (e.g., writer’s cramp or musician’s dystonia) and others are provoked by certain posture	A “geste antagoniste” (i.e., a sensory trick in which a simple, tactile, nonforceful action is directed toward the affected body part)	Observation of the abnormal movements or postures
Paroxysmal kinesigenic dyskinesia	Abrupt changes in movement Anxiety Stress Startle response	Aging	Observation of the abnormal movements or postures
Stiff-person syndrome	Sudden voluntary movement Physical touch Cold temperature Emotional upset Startle response	Use of benzodiazepines	Palpable rigidity of paraspinal and abdominal muscles Simultaneous contraction of agonist and antagonist muscle pairs Lumbar hyperlordosis Exaggerated startle response Increased deep-tendon reflexes

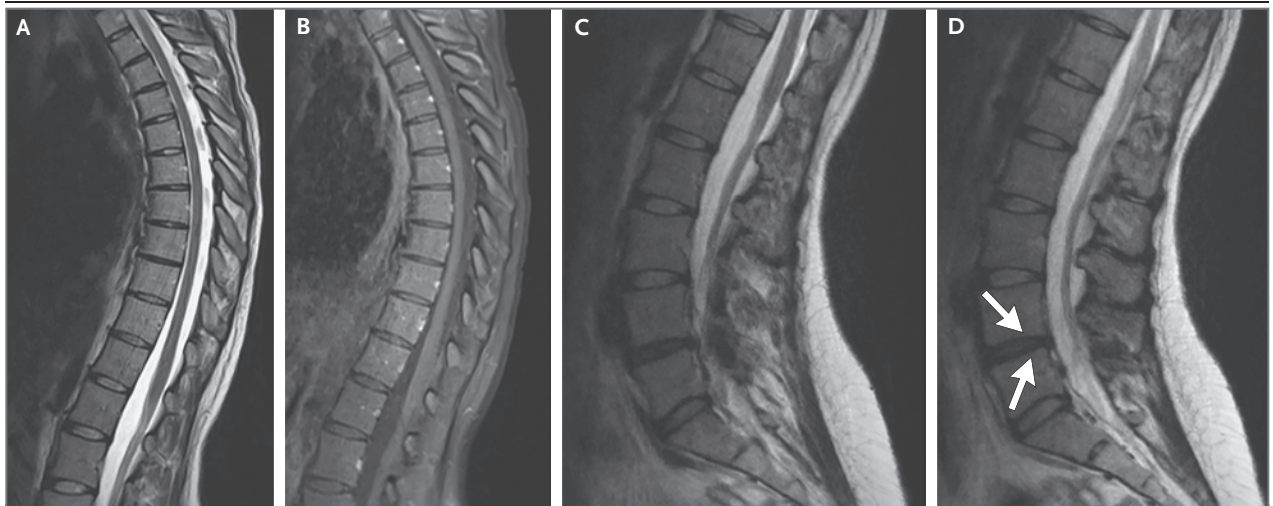


Figure 1. MRI of the Spine.

A sagittal T2-weighted image (Panel A), obtained without the administration of intravenous contrast material, and a sagittal gadolinium-enhanced, T1-weighted image with fat suppression (Panel B) show that the thoracic spine and spinal cord have a normal appearance. There is no abnormal enhancement. Sagittal T2-weighted images of the lumbar spine (Panels C and D), obtained without the administration of intravenous contrast material, show that the conus medullaris and cauda equina have a normal appearance (Panel C) and that mild degenerative disk disease is present at L4–L5 (Panel D, arrows) without radiographically significant spinal canal stenosis. Exaggeration of the normal lumbar lordosis can be seen.

involuntary muscle twitching (fasciculations or myokymia), which were absent in this patient.

CENTRAL NERVOUS SYSTEM DISORDER

Stiffness is more common with central nervous system disorders than with peripheral nervous system disorders. Spasticity is a disorder that is characterized by increased muscle tone due to upper motor neuron dysfunction.⁷ On examination, patients with spasticity have increased passive resistance with initiation of movement or during high-velocity movement; slow, continuous movement through the range of motion is met with some, albeit less, increased resistance. This patient was evaluated by multiple physicians over the course of her illness, but she was noted to have increased muscle tone on only one examination. The severity of spasticity is typically consistent across examinations.

Rigidity refers to a consistent increase in muscle tone, irrespective of the velocity of movement. Rigidity is commonly seen in patients with extrapyramidal disorders such as Parkinson's disease. However, this patient did not have any other parkinsonian features, such as tremor, bradykinesia, or postural instability. Neither spasticity nor rigidity fit well with this patient's presentation. One unusual feature of this patient's

clinical history was fluctuating and inconsistent symptoms across examinations. What neurologic disorders could cause episodic or fluctuating muscle stiffness?

DYSTONIA

Dystonia describes sustained or intermittent muscle contractions that cause abnormal movements and postures that are often repetitive.⁸ These contractions can be uncomfortable and can restrict voluntary movement, and they may be described by some patients as spasms. The muscle contractions are often triggered by voluntary movement, by a specific task (e.g., writing, resulting in writer's cramp), or by the sustaining of a specific posture. They are often relieved by a "geste antagoniste," or sensory trick — a simple, tactile, nonforceful action. For example, a patient with truncal dystonia could press the corner of a chair into a specific portion of the back. Although symptoms can fluctuate, dystonic movements are usually stereotyped and follow a specific pattern. This patient did not describe repetitive muscle contractions, and no sensory tricks relieved her symptoms. Moreover, truncal dystonia is quite rare.^{9,10}

Patients with paroxysmal dyskinesia have episodic bouts of involuntary dystonic or choreiform

movements.^{11,12} In patients with paroxysmal kinesigenic dyskinesia, episodes are triggered by abrupt changes in movement, as were seen in this patient. However, the typical age at onset is between 5 and 15 years, the typical duration of an episode is less than 1 minute, episodes are often preceded by a sensory aura, and patients often have up to 20 episodes per day — none of which fit with this patient's presentation.

HYPEREKPLEXIA

An important clue to the patient's diagnosis may be found in the description of her falls. Her legs stiffened in response to an unexpected physical stimulus — after unexpectedly striking her leg on a coffee table, she was unable to control her leg movements and could not protect herself from falling and striking her face on the floor. Hyperkplexia is a group of genetic disorders that is characterized by an exaggerated myoclonic startle response, although rare acquired forms have been described.¹³ The startle response is followed by a period of generalized stiffness that inhibits voluntary muscle control and can lead to a fall, as was the case in this patient. An exaggerated startle response was reported on examination of this patient. However, symptoms in patients with genetic hyperkplexia begin in infancy.

STIFF-PERSON SYNDROME

Stiff-person syndrome is an adult-onset, acquired autoimmune disorder that is associated with stiffness, gait dysfunction with falls, and an exaggerated startle response.¹⁴⁻¹⁷ Symptoms result from impaired γ -aminobutyric acid (GABA)-mediated inhibition of alpha motor neurons in the spinal cord and brain, which leads to hyperactivity. Stiffness and painful muscle spasms are the primary symptoms. In patients with classic stiff-person syndrome, stiffness begins in the paraspinal and abdominal muscles and progresses to involve the proximal legs. Sudden voluntary movement, physical touch, cold temperature, emotional upset, and startle response can all provoke painful spasms, which often occur in clusters. The combination of stiffness and superimposed spasms can lead to falls during which persons are unable to brace themselves.

The median age at onset of stiff-person syndrome is between 35 and 40 years, and this patient's symptoms started when she was 27 years of age. More than 50% of persons with stiff-person syndrome have a history of other autoimmune conditions, with up to 30% having type 1 diabetes mellitus. This patient had a history of vitiligo, eczema, alopecia areata, Graves' disease, and immune thrombocytopenia.

On examination, patients with classic stiff-person syndrome do not have weakness or sensory loss, and testing for increased muscle tone in the arms may show normal findings. Seventy percent of patients with stiff-person syndrome have increased deep-tendon reflexes, as this patient did. In addition to the exaggerated startle response noted on examination of this patient, several other examination maneuvers could help support a diagnosis of stiff-person syndrome in this patient.¹⁸ Palpation of the abdominal and paraspinal muscles may reveal hypertrophy and firmness. Paraspinal muscle rigidity may lead to lumbar hyperlordosis, a finding that was noted on this patient's two spinal MRI studies. Another hallmark of stiff-person syndrome is simultaneous contraction of agonist and antagonist muscle pairs that can limit range of motion. For example, engaging in active knee flexion may lead to counterproductive cocontraction of the knee extensors.

Clinical response to treatment with a benzodiazepine — typically diazepam — is another key diagnostic feature of stiff-person syndrome that was seen in this patient. However, this feature of stiff-person syndrome can lead to diagnostic delay if it is misunderstood. Patients with stiff-person syndrome frequently have coexisting anxiety, depression, or agoraphobia; the presence of such conditions, together with relief of symptoms after benzodiazepine therapy, may lead to misattribution of symptoms to psychiatric causes.¹⁹⁻²¹

Stiff-person syndrome is also frequently mistaken for functional neurologic disorder, and vice versa.^{22,23} Experts in functional neurologic disorder have emphasized the use of positive examination findings, including variability and distractibility of findings, to "rule in" a diagnosis of functional neurologic disorder.²⁴ The variability and reported distractibility that were observed with respect to stiffness in this patient

could easily be mistaken as signs of functional neurologic disorder. As compared with patients with functional neurologic disorder, patients with stiff-person syndrome more frequently have an exaggerated startle response, unexplained and injurious falls, associated systemic autoimmunity, and hyperreflexia — all of which were seen in this patient.²³

Identification of specific, consistent triggers that can explain the variability in this patient's symptoms is key to making an accurate diagnosis. Startle response or unexpected tactile stimuli seemed to consistently lead to spasm and worsening stiffness in this patient, which would make stiff-person syndrome a more likely diagnosis than functional neurologic disorder.

To confirm the diagnosis in this patient, I would recommend testing for glutamic acid decarboxylase 65 (GAD65) autoantibodies, which are detected in 60 to 90% of patients with classic stiff-person syndrome.^{14,15,23,25}

DR. CHRISTOPHER T. DOUGHTY'S DIAGNOSIS

Stiff-person syndrome.

DIAGNOSTIC TESTING

Dr. Jenny J. Linnoila: GAD65 autoantibodies are commonly found in patients with type 1 diabetes mellitus, autoimmune thyroiditis, and pernicious anemia, although typically at lower titers than those seen in patients with an autoimmune neurologic disease, such as stiff-person syndrome, epilepsy, limbic encephalitis, or cerebellar degeneration.²⁶ GAD65 autoantibodies can be detected by means of a radioimmunoassay, an enzyme-linked immunosorbent assay (ELISA), or a tissue-based indirect immunofluorescence assay. Although the tissue-based indirect immunofluorescence assay can be used to identify the presence of GAD65 autoantibodies (which bind to brain tissue), radioimmunoassays and ELISAs are most often used to quantify the antibody titer. The ELISA method can produce values that are orders of magnitude higher than those produced with the radioimmunoassay method, especially if the ELISA samples are diluted first. Therefore, when determining the potential clinical relevance of GAD65 autoantibodies, it is important to know the testing method that was

used, the GAD65 autoantibody titer of the patient's sample, and the typical ranges that are reported by the testing laboratory.

This patient's serum was tested for the presence of GAD65 autoantibodies with the use of a radioimmunoassay. The level of GAD65 autoantibodies was found to be 169 nmol per liter. The radioimmunoassay that was used for this patient cites a value of 20 nmol per liter as a cutoff to indicate neurologic autoimmune disease.²⁷ The patient's radioimmunoassay value of 169 nmol per liter was within the typical range for patients with GAD65 autoantibody-associated stiff-person syndrome whose samples were tested in the same laboratory.

LABORATORY DIAGNOSIS

Glutamic acid decarboxylase 65 autoantibody-associated stiff-person syndrome.

DISCUSSION OF MANAGEMENT

Dr. Kate Brizzi: If testing for GAD65 autoantibodies had been negative in this patient, I would have considered testing for autoantibodies against glycine receptor, amphiphysin, or dipeptidyl-peptidase-like protein 6, given her clinical presentation, which was characteristic of stiff-person syndrome. In addition, electromyography could have been performed to assess for the presence of continuous motor-unit activity in the paraspinal muscles or the presence of simultaneous contraction of agonist and antagonist muscle pairs.²⁸

An important step in the care of this patient was to rule out other possible causes of her elevated GAD65 autoantibody level. The blood level of glycated hemoglobin was normal, which ruled out type 1 diabetes mellitus. Although paraneoplastic syndromes are thought to be relatively uncommon among patients with classic presentations of stiff-person syndrome, elevated levels of GAD65 autoantibodies have been associated with breast cancer, lymphoma, and thymoma.²⁹ In this patient, positron-emission tomography and computed tomography of the chest, abdomen, and pelvis revealed a splenic lesion with ¹⁸F-fluorodeoxyglucose avidity, as well as right axillary lymphadenopathy. She was evaluated by an oncology consultant, and an excisional lymph-node biopsy was recommended.

Pathological evaluation of the excisional biopsy specimen was consistent with reactive lymphoid hyperplasia. Two MRI studies of the abdomen that were performed 3 months apart showed that the size of the splenic lesion was decreasing, which made cancer an unlikely diagnosis.

A multidisciplinary approach is beneficial in the management of stiff-person syndrome. In this patient, physical therapy (including aqua therapy) had been recommended. Data to guide physical therapy approaches in patients with stiff-person syndrome are limited; the goals of therapy for this patient included flexibility, pain relief, and functional mobility. With regard to drug therapy, relief of symptoms can often occur after treatment with benzodiazepines or muscle relaxants, usually at high doses. In patients with severe symptoms or in those who have adverse side effects from drug therapies, immunotherapy with intravenous immune globulin may be beneficial.^{28,30,31} Anxiety or a fear of falling is typical of patients with stiff-person syndrome and can worsen the symptoms of the disease. Management includes medications, psychotherapy, and mindfulness-based approaches.

After a discussion with the patient about treatment options, the patient elected to start treatment with diazepam for relief of her symptoms, in addition to physical therapy and psychotherapy. At a follow-up visit 1 month later, she reported marked improvement in ambulation and in her ability to climb and descend stairs, as well as improvement in her ability to participate in physical activities. This improvement was sustained over the course of the next year with adjustments in the diazepam dose and in ongoing multidisciplinary care. Approximately 2 years after diagnosis, the patient began to have worsening of symptoms with an increased fear of falling. Treatment with intravenous immune globulin was initiated, and her symptoms abated.

PATIENT PERSPECTIVE

The Patient: My symptoms became suddenly unbearable shortly after my wedding. I missed my first day of work at a new job because I went upstairs and was unable to come down. My legs would not bend at the knee and instead became stiff, like they were made of wood. Several times, I managed to walk gingerly to the top of

the stairs, but as soon as I tried to go down, I froze again. I did not have my phone and my husband was out. I was stuck upstairs for more than 2 hours as my body was getting stiffer and stiffer, with the growing mental anguish of not knowing what was happening to me. I worried about missing work and what to offer as an excuse. Should I just vaguely say I was sick? That my back hurt? Who would believe me if I said I was stuck in my house for more than 2 hours because I was afraid of the stairs?

Receiving a diagnosis of stiff-person syndrome is terrifying because of its rarity and the uncertainty it brings. Learning to live with the uncertainty has been one of the most difficult tasks of my life. However, I am lucky to have a supportive, empathetic, and trustworthy neurologist, whose care has been instrumental in my learning to accept and live with this diagnosis. My husband, parents, sisters, and friends have also been very supportive. I am involved in the Stiff Person Syndrome Research Foundation, an organization working to increase awareness of, and find a cure for, stiff-person syndrome. On a personal level, it offers me a supportive and safe space. I also have a cat, whose support and loyalty are questionable, but he's been a consistent source of levity and laughter.

I still have bad days when I am terrified and worry about my future. I use psychotherapy and practice meditation, and I try to find creative outlets to keep my mind from spiraling. I have also taken stress management and resiliency training.

I am grateful that I can resolve most everyday symptoms with medications. But in higher-stakes situations where I have less control, I need more medication, and even then, symptoms may emerge. I have started to avoid situations that are more likely to trigger symptoms. This limits my world and is a reason why I desperately want better treatment options and a cure for stiff-person syndrome.

FINAL DIAGNOSIS

Glutamic acid decarboxylase 65 autoantibody-associated stiff-person syndrome.

This case was presented at Neurology Grand Rounds.

Disclosure forms provided by the authors are available with the full text of this article at NEJM.org.

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